

**CLUSTER FOR CLEAN ENERGY AND
CLIMATE CHANGE (CCECC), BOWEN
UNIVERSITY, IWO.**



**1ST NATIONAL CONFERENCE ON CLEAN
ENERGY AND CLIMATE CHANGE
15-16TH APRIL, 2026**

C C E C C P R O C E E D I N G S

**Advancing Clean Energy
Solutions for Climate-
Resilient and Sustainable
Development in Nigeria**

Buoyancy-driven heat transfer enhancement in a hexagonal cavity using an optimized MgO–CuO–SiC hybrid nanofluid

Olabemiwo J. T., Okekunle P. O.

Department of Mechanical Engineering, Ladoke Akintola University of Technology, Nigeria

*Corresponding Author: jekayinoluwa.olabemiwo15@kwasu.edu.ng

Abstract

Hybrid nanofluids possess synergistic effects capable of enhancing buoyancy-driven heat transfer in engineering systems that depend on natural convection. However, the relationship between nanoparticle composition, thermophysical properties, and cavity geometry is nonlinear. This makes it difficult to reliably predict thermal performance. This study investigated the properties of ternary hybrid nanofluids under natural convection conditions in a hexagonal cavity with a focus on the impact of nanoparticle composition and geometric parameters on heat transfer and fluid flow for passive thermal management applications. This study used a ternary nanofluid composed of Magnesium oxide (MgO), Copper oxide (CuO), and Silicon carbide (SiC) dispersed in a base fluid of ethylene glycol. A mixture-based Response Surface Methodology (RSM) implemented in Design-Expert (v13.0) was used to generate five (5) optimal ternary compositions with nanoparticle volume fractions between 0–4% and base fluid content between 96–99%. The optimal ternary nanofluid composition consisted of 0.58612% MgO, 0.64405% CuO, 2.58917% SiC, and 96.18067% ethylene glycol. For each composition, effective thermophysical properties such as thermal conductivity (k), specific heat (C_p), and dynamic viscosity (μ) were computed using established theoretical models. The average k , C_p , μ , and ρ were 0.2742885 W/m·K, 2279.17221 J/kg·K, 0.0168881 Pa·s, and 1184.22594 kg/m³, respectively. The laminar natural convection governing equations were expressed in dimensionless form and solved using the Finite Element Method (FEM) in COMSOL Multiphysics (v5.6). The influence of inclination angle θ was investigated over Rayleigh numbers, with a fixed aspect ratio (AR) of 0.25. At $Ra = 10^2$ the average Nusselt numbers (Nu) were 2.760979, 2.760986, 2.767196, 2.770342, and 2.760979 for $\theta = 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$, respectively. At $Ra = 10^4$, the Nusselt numbers increased to 3.147658, 3.139457, 3.140626, 3.145446, and 3.147658. At $Ra = 10^5$, the values were 4.680454, 4.689514, 4.757156, 5.035523, and 4.680454. This indicates a sensitivity to inclination under convection conditions.

Keywords

Nanofluid, Natural Convection, Hexagon, Thermal Conductivity, Rayleigh Number, Nusselt Number

1. Introduction

Natural convection is a heat transfer process that causes fluid motion without an external force. Temperature gradient causes a density difference, which results in the fluid motion

(Salahuddin et al., 2026). Natural convection is an active area of research due to its application in solar collectors, HVAC (heating, ventilation, and air conditioning) systems, thermal insulation and electronics cooling (Hasan et al., 2025). A nanofluid consists of suspensions of nanoparticles smaller than 100nm in a base fluid. Hybrid nanofluids have multiple nanoparticles. Recently, the efficiency of hybrid nanofluids for the enhancement of natural convection has been studied for applications in photovoltaic-thermal (PVT) systems, automobile radiators and industrial heat exchangers. Hence, the use of hybrid nanofluids in systems that depend on natural convection has potential for thermal performance, especially in enclosures where internal natural convection occurs (Farooq et al., 2025). However, the relationship between the composition of nanoparticles in the hybrid nanofluid mixture and the resultant thermophysical properties is nonlinear (Farooq et al., 2025). This study employed response surface methodology in studying the relationship between the nanoparticle composition, nanofluid thermophysical properties, and the geometric parameters of the cavity under natural convection conditions.

2. Literature Review

Mono-nanofluids consisting of single nanoparticles like Al_2O_3 and CuO have been proven to provide improved thermal performance in natural convection systems, but their extensive practicability is limited as they agglomerate or sediment at high temperature or after extensive usage (Ibeh et al., 2021). Zemani *et al.* (2023) used the SIMPLE algorithm to simulate the natural convection in a U-shaped enclosure fitted with a T-shaped baffle and filled with CuO -water nanofluid. Their results showed that a rise in Rayleigh number, aspect ratio, or nanoparticle volume fraction increases the mean Nusselt number, which implies enhanced heat transfer performance. Hybrid nanofluids, however, give higher thermal conductivity with better stability due to the synergistic effects between the dissimilar constituent nanoparticles (Zainon and Azmi, 2021). Ternary hybrid nanofluids are hybrid nanofluids with three types of nanoparticles. The combination of the three nanoparticles produces a synergistic thermal effect that gives improved heat transfer and nanoparticle dispersion in the nanofluid (Ullah et al., 2025). Omri et al. (2022) considered the impact of the power law index applied to a hybrid nanofluid cavity. They found out that the average Nusselt number increased with increasing Rayleigh number, with the peak at $Ra=106$. They also reported that increasing the ratio of thermal conductivity gave an increase in heat transfer rate at the heated boundary and the solid-fluid layer. Research also shows that geometric complexity tends to aid heat transfer enhancement. Using complex enclosure shapes or adding inner obstacles to cavities improved heat transfer and fluid flow. Saboj *et al.* (2023) analyzed the entropy production in an octagonal cavity with an inner cold cylinder. Their findings indicated that the inner cold cylinder aided heat transfer, and further heat transfer improvement occurred with the addition of the water-based aluminum-oxide nanofluid. Most natural convection studies focused on mono- or binary hybrid nanofluids in simple cavity shapes. This has left ternary hybrid nanofluids unexplored for their impact on natural convection. Research on complex cavities, like polygonal-shaped enclosures, is also limited. This study addressed these gaps by exploring the influence of a ternary hybrid nanofluid in a hexagonal cavity in a bid to determine the optimal heat transfer conditions.

3. Methodology

3.1 Selection of Base Fluid and Nanoparticles

Magnesium oxide (MgO), Copper oxide (CuO) and Silicon carbide (SiC) nanoparticles were selected for this study. MgO possesses high thermal conductivity, is relatively stable chemically and is cost-saving (Herng et al., 2023). CuO has excellent thermal conductivity and stability (Karami et al., 2016). SiC has very high thermal conductivity and stability, which makes it a perfect candidate for extreme temperature conditions (Khan *et al.*, 2024). Ethylene glycol (EG) was used as the base fluid. It has a wide operating temperature range due to its high boiling and freezing points. It also has low volatility and is compatible with several nanoparticles (Ahmad et al., 2025).

3.2 Formulation of Ternary Hybrid Nanoparticle Percentage Composition

The response surface methodology approach used to determine the optimal composition for the MgO–CuO–SiC/ethylene glycol nanofluid was a mixture design. The Design-Expert version 13 software was used for the design. In the design, the volume fractions of the nanoparticles were varied within 0-4% of the nanofluid mixture, while the base fluid (ethylene glycol) covered the rest of the mix. Studies have shown that extending the nanoparticle composition beyond 4% of the nanofluid mix could result in more viscosity or sedimentation, which could make the nanofluid lose the stability of nanoparticle dispersion (Bacha *et al.*, 2024).

3.3 Estimation of Thermophysical Properties

The effective thermal conductivity (k_{tnf}), effective specific heat ($C_{p_{tnf}}$), effective dynamic viscosity (μ_{tnf}), effective density (ρ_{tnf}), effective thermal diffusivity (α_{tnf}), and effective thermal expansion coefficient (β_{tnf}) of the nanofluid were estimated with the following equations 1–6 in agreement with Kahveci and Ögüt (2020) and Islam et al. (2023). They were used to estimate the thermophysical properties of the ternary hybrid nanofluid compositions derived in the study.

$$k_{tnf} = K_{bf} \left[\frac{(k_h + 2K_{bf}) - 2\varphi_{tnf}(K_{bf} - K_h)}{(k_h + 2K_{bf}) + \varphi_{tnf}(K_{bf} - K_h)} \right] \quad 1$$

$$C_{p_{tnf}} = \frac{(1 - \varphi_{tnf})\rho_{bf}C_{p_{bf}} + \sum_{i=1}^n \varphi_i \rho_i C_{p_i}}{\rho_{tnf}} \quad 2$$

$$\mu_{tnf} = \frac{\mu_{bf}}{(1 - \varphi_1)^{2.5}(1 - \varphi_2)^{2.5}(1 - \varphi_3)^{2.5}} \quad 3$$

$$\rho_{tnf} = (1 - \varphi_{tnf})\rho_{bf} + \sum_{i=1}^n \varphi_i \rho_i \quad 4$$

$$\alpha_{tnf} = \frac{K_{tnf}}{(\rho_{tnf} C_{p_{tnf}})} \quad 5$$

$$\beta_{tnf} = \frac{(1 - \varphi_{tnf})\beta_{bf}\rho_{bf} + \sum_{i=1}^n \varphi_i\beta_i\rho_i}{\rho_{tnf}} \quad 6$$

The total volume fraction of the ternary hybrid nanofluid (φ_{tnf}) is the combined concentration of all the nanoparticles suspended in the base fluid. It was estimated according to equation 7 (Kahveci and Ögüt, 2020; Islam *et al.*, 2023) given as:

$$\varphi_{tnf} = \varphi_1 + \varphi_2 + \varphi_3 + \dots + \varphi_n \quad 7$$

3.4 Determination of Optimal Nanofluid Composition

The optimization procedure implemented in Design -Expert was multi-objective to obtain the most thermally efficient ternary nanoparticle composition within the prescribed mixture constraints. The optimization focused on maximizing the effective thermal conductivity (k_{tnf}), and effective specific heat capacity ($C_{p,tnf}$) while minimizing the dynamic viscosity (μ_{tnf}) within acceptable flow limits, while maintaining all nanoparticle volume fractions within 0–0.04 of the total nanofluid mix. Further applying a desirability-based approach gave multiple feasible solutions with a desirability score of 1.0 (unity) out of which three (3) optimal compositions that offered the best combination of thermophysical properties were selected as shown in Table 1. The selection criteria favoured high predicted thermal conductivity, specific heat capacity exceeding 2200 J/kg·K, moderate viscosity values and the three nanoparticles being represented in the composition. The selected optimized compositions were then implemented in COMSOL Multiphysics 5.0 as defined material property inputs for heat transfer and fluid flow analysis.

Table 1: Selected Optimized Ternary Hybrid Nanoparticle Compositions for Numerical Simulation

MgO	CuO	SiC	K_{tnf} W/m·K	$C_{p,tnf}$ J/kg·K	μ_{tnf} Pa·s
0.0131739	0.00078651	0.0133976	0.273068374	2286.148794	0.016818875
0.0261928	0.00154981	0.00434146	0.276728806	2265.219028	0.01702649
0.0131739	0.00078651	0.0133976	0.273068374	2286.148794	0.016818875

3.5 Development of Numerical Model

Equations 8–11 are the dimensional forms of the Navier-Stokes and energy equations modeling the heat transfer and fluid flow.

Continuity Equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \text{Islam et al. (2023)} \quad 8$$

X-Momentum Equation (Navier-Stokes):

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad \text{Islam et al. (2023)} \quad 9$$

Y-Momentum Equation (Navier-Stokes):

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial P}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + g\beta(T - T_0) \quad \text{Islam et al. (2023)} \quad 10$$

Energy Transport Equation:

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha_{nf} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \quad \text{Islam et al. (2023)} \quad 11$$

The parameters in equations 12 and 13 as follows were applied for normalization:

$$X = \frac{x}{L}, \quad Y = \frac{y}{L}, \quad U = \frac{uL}{\nu_{nf}}, \quad V = \frac{vL}{\nu_{nf}}, \quad \theta = \frac{T - T_c}{T_h - T_c} \quad \text{Ambethkar (2023)} \quad 12$$

$$Ra = \frac{g\beta(T_h - T_c)L^3}{\nu_{nf}^2}, \quad Pr = \frac{\nu_{nf}}{\alpha_{nf}}, \quad \alpha_{nf} = \frac{k_{nf}}{\rho_{nf}C_{p,nf}} \quad \text{Ambethkar (2023)} \quad 13$$

Using the transformation parameters from equations 5 and 6, we get the dimensionless equations in 14–17:

Continuity:

$$\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y} = 0 \quad \text{Ambethkar (2023)} \quad 14$$

X-Momentum:

$$\frac{\partial U}{\partial t} + U \frac{\partial U}{\partial X} + V \frac{\partial U}{\partial Y} = -\frac{\partial P}{\partial X} + \frac{1}{Re} \left(\frac{\partial^2 U}{\partial X^2} + \frac{\partial^2 U}{\partial Y^2} \right) \quad \text{Ambethkar (2023)} \quad 15$$

Y-Momentum:

$$\frac{\partial V}{\partial t} + U \frac{\partial V}{\partial X} + V \frac{\partial V}{\partial Y} = -\frac{\partial P}{\partial Y} + \frac{1}{Re} \left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2} \right) + RaPr\theta \quad \text{Ambethkar (2023)} \quad 16$$

Energy:

$$\frac{\partial \theta}{\partial t} + U \frac{\partial \theta}{\partial X} + V \frac{\partial \theta}{\partial Y} = \frac{1}{RePr} \left(\frac{\partial^2 \theta}{\partial X^2} + \frac{\partial^2 \theta}{\partial Y^2} \right) \quad \text{Ambethkar (2023)} \quad 17$$

The momentum equations defined the fluid flow across the X and Y axis while the energy equation defined the temperature field. The Rayleigh number (Ra) defined the buoyancy strength of the flow. The dimensionless thermophysical properties obtained from equations 1-6 were therefore substituted into the dimensionless equations 14-17 to govern the fluid flow and heat transfer simulation. Nusselt number (Nu), which is the output variable of the combination of the equations in the simulation, is a function of the temperature gradient (θ).

The geometry under consideration was a two-dimensional solid hexagon cavity with a square obstacle in its middle. The left-bottom and horizontal bottom walls of the cavity were heated. The remaining walls were assumed to be insulated and adiabatic. The four walls of the inner square obstacle were kept at a cold temperature. The grid-sensitive mesh contained 15690 non-uniform elements at the resolution shown in Figure 1. The Finite Element Method (FEM) was used to develop the numerical model and simulate the fluid flow and heat transfer based on the boundary conditions.

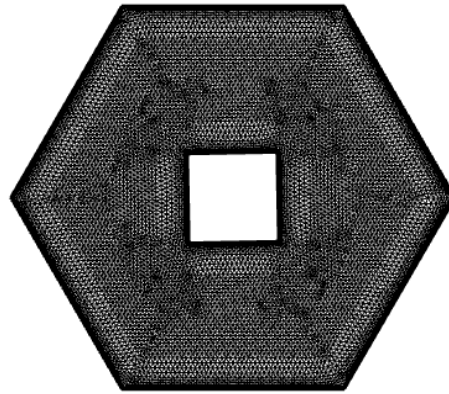


Figure 1: Diagram of the refined mesh of the computational domain

4. Results

To validate the numerical implementation, the present model was benchmarked against established solutions for natural convection in a differentially heated square cavity. The average Nusselt number agreed with studies of the same problem (de Vahl Davis (1983); Thilagavathi and Prasad (2025)) as shown in Table 2.

Table 2: Comparison of the average Nusselt number for the present study and published literature

Rayleigh Number (Ra)	Average Nusselt Number		
	10^3	10^4	10^5
de Vahl Davis (1983)	1.118	2.243	4.519
Thilagavathi & Prasad (2025)	1.1183	2.2431	4.5197
Present Study	1.11744	2.24338	4.51555

5. Discussion

Figure 2 shows the impact of different inclination angles ($\theta = 0^\circ, 30^\circ, 45^\circ, 60^\circ$, and 90°) in terms of the average Nusselt number of the inner square block at different Rayleigh numbers (Ra). The average Nusselt number was almost constant for all inclination angles until a higher Ra ($Ra = 10^4$) was reached, from which a sharp increase occurred due to the dominance of convection at that point. Ibrahim et al. (2022) also observed in their study that changing inclination angles had an insignificant influence at low Rayleigh numbers, but their effect became more pronounced at high Rayleigh numbers.

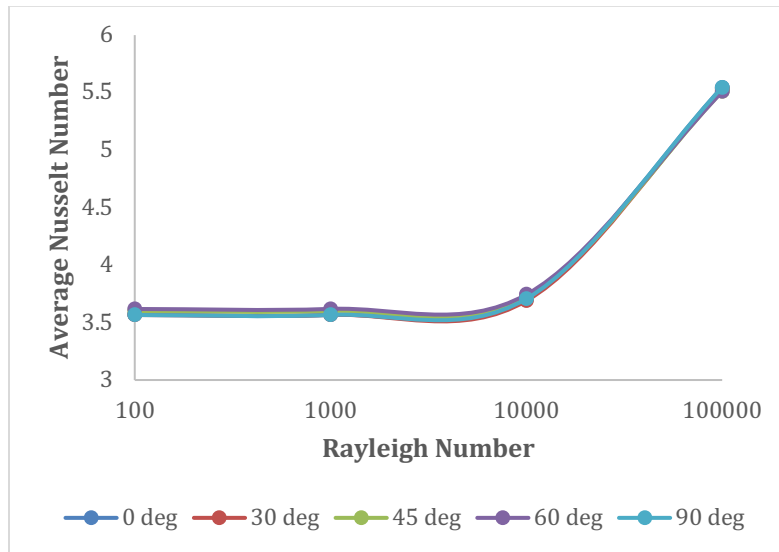


Figure 2: Effect of Angle of Inclination on Nusselt Number

Figure 3 shows the trend of the average Nusselt number for different aspect ratios ($AR = 0.25, 0.30, 0.35$, and 0.40) of the enclosure. The average Nusselt number generally increases as the aspect ratio increases, and the trend becomes more pronounced at high Rayleigh numbers when there are stronger circulation and buoyancy-driven flow.

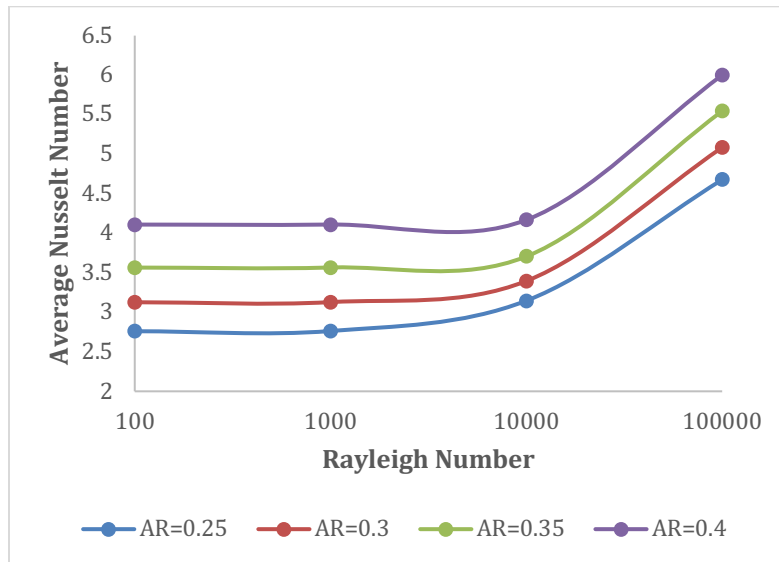


Figure 3: Effect of Aspect Ratio on Nusselt Number

6. Conclusion and Recommendations

The system considered is a steady-state natural convective flow in a hexagonal cavity with an inner square block. The cavity was filled with a ternary hybrid nanofluid, MgO-CuO-SiC/ethylene glycol. The hybrid nanofluid mix enhanced thermal conductivity and specific heat capacity while keeping viscosity balanced. High Rayleigh numbers gave strong buoyancy as convection dominated the flow. An increase in aspect ratio resulted in stronger circulation due to the occurrence of longer flow paths. Changing the inclination angle changed the flow symmetry, and higher inclination angles enhanced heat transfer. These results show that the ternary hybrid nanofluid composition improved thermal conductivity and maintained reasonable viscosity to not impede the flow. This is useful in cooling systems, passive thermal design and enclosures. This study recommends future investigation into other polygonal cavities and novel ternary hybrid nanofluids for the optimization of thermal efficiency.

Conflict of Interest

None

Funding Statement

Mention funding sources if applicable.

References (APA 7th Edition Format Example)

Ahmad, H., Guedri, K., Ahmad, S. et al. Numerical and machine learning based evaluation of ethylene glycol based hybrid nano-structured (TiO₂-SWCNTs) fluid flow. *Sci Rep* 15, 6084 (2025). <https://doi.org/10.1038/s41598-025-88789-5>

Ambethkar, V. (2023). Impact of the Nusselt and Rayleigh numbers on heat transfer in rectangular enclosure. *International Journal of Applied Mathematics*, 36(5): 651–661. <https://doi.org/10.12732/ijam.v36i5.5>

Bacha, H. B., Ullah, N., Hamid, A., and Shah, N. A. (2024). A comprehensive review on nanofluids: Synthesis, cutting-edge applications, and future prospects. *International Journal of Thermofluids*, 22, 100595. <https://doi.org/10.1016/j.ijft.2024.100595>

de Vahl Davis, G. (1983). Natural convection of air in a square cavity: A benchmark numerical solution, *International Journal for Numerical Methods in Fluids*, 3(3), pp. 249–264.

Farooq S., Habib M., Cardozo O., Ullah K., Pandey A.K., Said Z. (2025) Exploring the impact of particle stability, size, and morphology on nanofluid thermal conductivity: A comprehensive review for energy applications. *Advances in Colloid and Interface Science*, Volume 341(103495). ISSN 0001-8686. <https://doi.org/10.1016/j.cis.2025.103495>

Hasan A., Ali M.M, Hossain S., Siam N. H., Rony A., Shohan A. (2025). A comprehensive review on natural convection in various shaped enclosures by FEM: Engineering applications. *South African Journal of Chemical Engineering*, Volume 54, Pages 308-334. ISSN 1026-9185. <https://doi.org/10.1016/j.sajce.2025.08.009>

Herng, S. Y., Rosli, M. A. M., Salimen, N., Herawan, S. G., and Hussain, F. (2023). Performance evaluation of photovoltaic thermal using MgO nanofluid. *Journal of Advanced Research in Fluid Mechanics and Thermal Sciences*, 109(2): 184–209. <https://doi.org/10.37934/arfmts.109.2.184209>

Ibeh S., Ndaliman M.B, Ezedimbu U. Audu J., Anthony O.C., Osther E., Yusuf A.L., Lawrence U.C., Illiyasu A. (2021). Mono nanofluids: review on current status, challenges and future prospects. *International Journal of Research and Innovation in Applied Science*. Volume VI (III), page 101. <https://dx.doi.org/10.51584/IJRIAS.2021.6302>

Ibrahim, M.N.J., Hammoodi, K.A., Abdulsahib, A.D., Flayyih, M.A. (2022). Study of natural convection inside inclined nanofluid cavity with hot inner bodies (circular and ellipse cylinders). *International Journal of Heat and Technology*, Vol. 40, No. 3, pp. 699-705. <https://doi.org/10.18280/ijht.400306>

Islam, T., Alam, M. N., Niazai, S., Khan, I., Fayz-Al-Asad, M., and Alqahtani, S. (2023). Heat generation/absorption effect on natural convective heat transfer in a wavy triangular cavity filled with nanofluid. *Scientific Reports*, 13: Article 21171. <https://doi.org/10.1038/s41598-024-67493-w>

Kahveci, K., and Ögüt, E. B. (2020). Natural convection of ethylene glycol and water mixture based Al_2O_3 nanofluids between vertical concentric cylinders. *Journal of Thermal Science and Technology*, 40(2): 237–250. <https://doi.org/10.47480/isibted.817005>

Karami M., Akhavan-Behabadi M.A., Dehkordi M.R., Delfani S. (2016). Thermo-optical properties of copper oxide nanofluids for direct absorption of solar radiation. *Solar Energy Materials and Solar Cells*. Volume 144, pages 136-142. ISSN 0927-0248. <https://doi.org/10.1016/j.solmat.2015.08.018>

Khan, S. A., Khan, S. A., Rahman, O., Masood, A., Khan, A. A., Zuberi, M. U., Bakhsh, F. I., and Islam, M. R. (2024). Impact of silicon carbide, boron nitride, and zirconium dioxide nanoparticles on ester-based dielectric fluids. *IET Nanodielectrics*. <https://doi.org/10.1049/nde2.12090>

Omri, M., Jamal, M., Hussain, S., Kolsi, L., and Maatki, C. (2022). Conjugate natural convection of a hybrid nanofluid in a cavity filled with porous and non-Newtonian layers: The impact of the power law index. *Mathematics*, 10(12): 2044. <https://doi.org/10.3390/math10122044>

Saboj, J. H., Nag, P., Saha, G., and Saha, S. C. (2023). Entropy production analysis in an octagonal cavity with an inner cold cylinder: A thermodynamic aspect. *Energies*, 16(14): Article 5487. <https://doi.org/10.3390/en16145487>

Salahuddin T., Ehsan A., Awais M., Khan M., Muhammad S., and Idrees M. (2026). Heat transfer characteristics and variable thermophysical properties of naturally convective Flow of viscous fluid inside a porous square cavity. *Heat Transfer*, Volume 1(55), 314-331. <https://doi.org/10.1002/htj.70077>

Thilagavathi, and Prasad, V. R. (2025). Thermal heat transfer enhancement analysis of magnetic ternary hybrid nanofluid in a hexagonal cavity with square obstacle. *Journal of Applied and Computational Mechanics*, 11(1): 239-252. <https://doi.org/10.22055/jacm.2024.46544.4548>

Ullah, M.E., Idrees, M., Saeed, S.T., & Zubaidi, A.A. (2025). Prandtl ternary nanofluid flow with magnetohydrodynamics and thermal effects over a 3D stretching surface using convective boundary conditions. *ZAMM - Journal of Applied Mathematics and Mechanics*, page 105. <https://doi.org/10.1002/zamm.70078>

Zainon, S. N. M., and Azmi, W. H. (2021). Recent Progress on Stability and Thermo-Physical Properties of Mono and Hybrid towards Green Nanofluids. *Micromachines*, 12(2), 176. <https://doi.org/10.3390/mi12020176>

Zemani, F., Ladjedel, O., and Sabeur, A. (2023). Simulation of CuO-water nanofluid natural convection in a U-shaped enclosure with a T-shaped baffle. *Journal of Engineering and Applied Science*, 70: Article 99. <https://doi.org/10.1186/s44147-023-00257-x>